

एन एस आई सी
N S I C

NSIC - TECHNICAL SERVICES CENTRE

रा. ल. उ. नि. तकनीकी सेवा केन्द्र

NATIONAL SMALL INDUSTRIES CORPORATION LIMITED

राष्ट्रीय लघु उद्योग निगम लिमिटेड

(A Govt. of India Enterprise / भारत सरकार का उपक्रम)

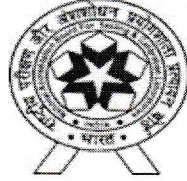
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TEST REPORT / परीक्षण रिपोर्ट



TC 7544

Diesel Engine Testing Laboratory

Constant Speed Compression Ignition Engine for Agricultural
Purpose Test Report as per IS 11170 : 1985

PERFORMANCE TEST

Report No.	:	NTSC-RJT/DETL/2025030
Job Card No.	:	2566
Date of Report	:	03-02-2025
ULR No.	:	TC754425100000030P
Testing Commencement	:	28-01-2025
Testing Completion	:	29-01-2025

TEST REPORT AS PER IS: 11170 : 1985
ULR No. TC754425100000030P
REPORT NO: NTSC-RJT/DETL/2025030

DATED : 03-02-2025

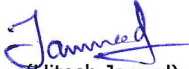
PART A : PARTICULARS OF SAMPLE SUBMITTED

a) NATURE OF SAMPLE : Single Cylinder,Vertical,Air Cooled,Constant Speed Compression Ignition
b) GRADE/VARIETY/TYPE/ : Diesel Engine for Agricultural Purposes, Governing class - B1
CLASS/SIZE ETC.
c) BRAND NAME, IF ANY : Model No. - MRS 5103 (MARIN APPLICATION)
d) DECLARED VALUES : POWER SPEED SFC
IF ANY (HP/kW) (RPM) (gm/kwh)
10/7.35 3000 300
e) SAMPLE CODE NO. : -
f) QR CODE NO. : -
g) BATCH NO. AND : Sr. Engine No.- V24 L 100096
DATE OF MANUFACTURE
h) QUANTITY : One
i) MODE OF PACKING : -
j) DATE OF RECEIPT : 17-01-2025
k) BIS SEAL : -
l) IO'S SIGNATURE : -
m) ANY OTHER INFORMATION : BIS Ref No: - Dated:
Date of test: 28-01-2025
Engine Manufacture By : VISHWAKARMA INDUSTRIES

NEW SERVEY NO.3161, MODH GNATI NI
VADI, OLD SANJAY HOLL PASE,
BOTAD - 364710

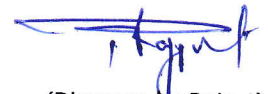
NOTE:1) This report, in full or in part, shall not be published, advertised,used for any legal action unless
prior permission has been secured from the The Centre Head of NSIC - Technical Services Centre, Rajkot.

2) This test report is ONLY FOR THE SAMPLE TESTED


(Hitesh Jamod)
Prepared By




(Raviprakash Vala)
Checked By

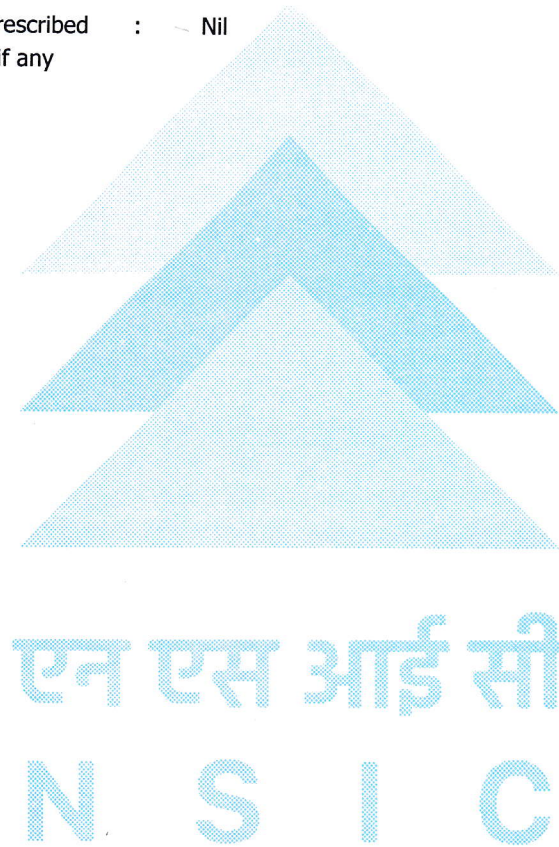

(Dharmendra Rajput)
Authorized Signatory

REPORT NO: NTSC-RJT/DETL/2025030

IS: 11170 : 1985

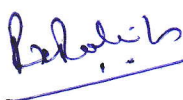
PART B: SUPPLEMENTARY INFORMATIONS

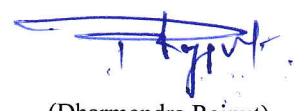
- a) Reference to sampling procedure, : -
Wherever applicable
- b) Supporting documents for the measurements : Governing test : Annexure-I
taken and results derived like graphs, Graphs : Annexure-II
tables, sketches, and/or photographs as Rating test & Loop Test : Annexure-III
appropriate to test report, if any
- c) Deviation from test methods as prescribed : Nil
in relevant ISS/work instructions, if any




(Hitesh Jamod)
Prepared by




(Raviprakash Vala)
Checked by


(Dharmendra Rajput)
Authorized Signatory

REPORT NO :
PART C. TEST RESULTS:

NTSC-RJT/DETL/2025030

IS : 11170 : 1985

Clause No.	Name Of Test	Test Method	Requirements as per specifications	Results
4.1	<u>i) RATING TEST:</u> POWER OUTPUT i) At site condition ii) Corrected at standard operating condition Overload Capacity Temperature rise of Lub.Oil Temperature rise of the crank case	(IS 10000:1980) PART-8 (IS 10000:1980) PART-4 Clause No.- 5.0	Power Output shall not be less than 1 percent of the declared value. 7.35 KW 7.35 KW } Engine Shall be capable to deliver an output of 10% in excess of its rated output at its rated speed without undue heating of the engine or any other mechanical trouble. There shall be no undue heating of the engine or any other mechanical trouble. No undue heating or any other mechanical trouble	7.31 KW 7.36 KW 8.09 KW (Withstood) No undue heating No undue heating
4.2	<u>ii) FUEL CONSUMPTION TEST</u> a) Specific fuel consumption at rated output in Gm/Kwhr Max i) At site condition ii) Corrected at standard operating condition b) Fuel Consumption at no load	(IS 10000:1980) PART-8 Clause No.- 2.1.3(b) (IS 10000:1980) PART-4 Clause No.- 3.1	Value as per IS : 309.0 (Gm/Kwhr) 309.0 (Gm/Kwhr) A tolerance of ± 5 percent in fuel consumption at 100 percent of the rated load shall be allowed unless otherwise agreed to between the manufacturer and the purchaser.	301.2 Gm/Kwh 300.9 Gm/Kwh Satisfactory 686.5 Gm/Hr
4.3	<u>iii) LUBRICATING OIL CONSUMPTION</u>	(IS 11170:1985)	Lubrication oil consumption shall not exceed declared value by more than 10 percent. 1% of SFC ltr/hr	0.014 ltr/hr
4.4	<u>iv) GOVERNING TEST (CLASS B₁)</u> a) Maximum change of speed as percentage of rated speed due to the following load variation : On suddenly taking off rated load I) Temporary change II) Permanent change b) Steady load speed band when the engine is delivering between 10% & 100% Of the rated power output. c) External means for adjusting the nominal speed of engine.	(IS 10000:1980) PART-7 Clause No.- 2.3	Max 15 % Max 8 % 1 %	10.07 % } Ref. 6.03 % } (Annexure-I) Satisfactory Provided

PART D. REMARKS:

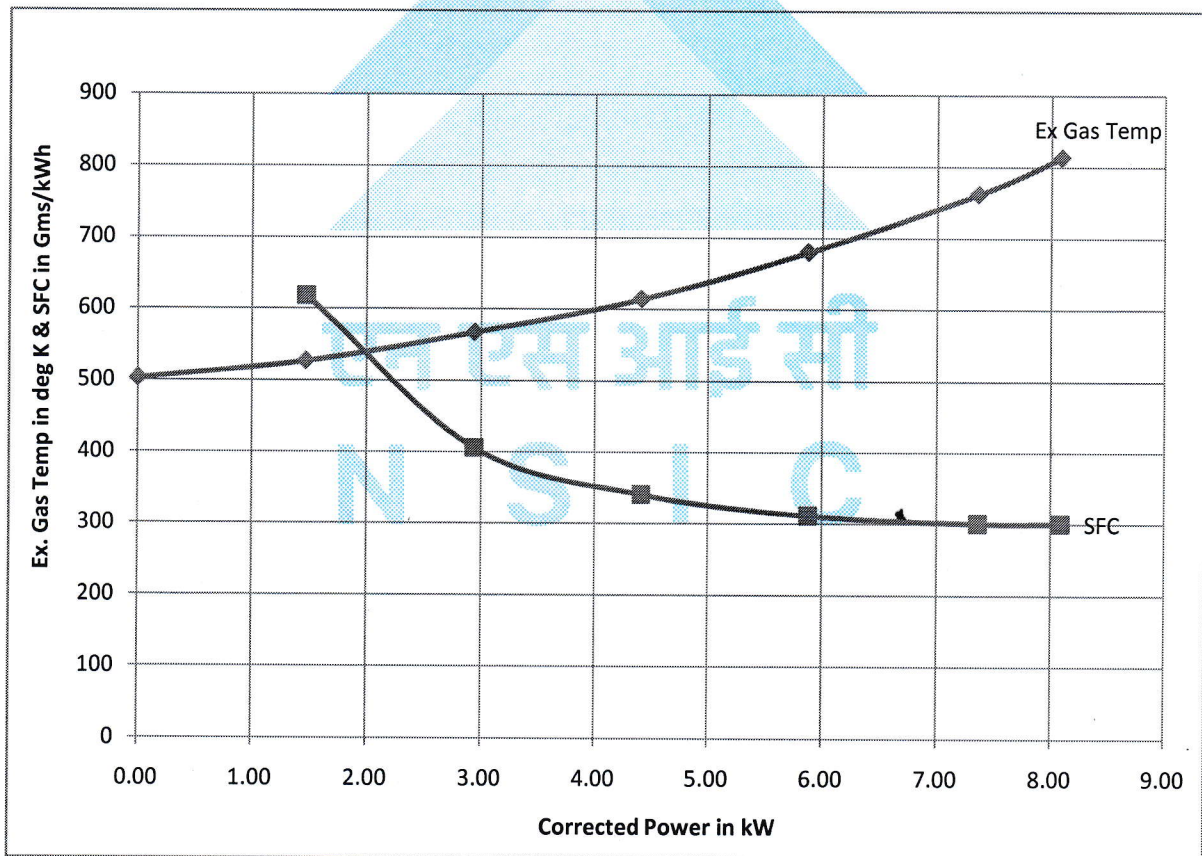
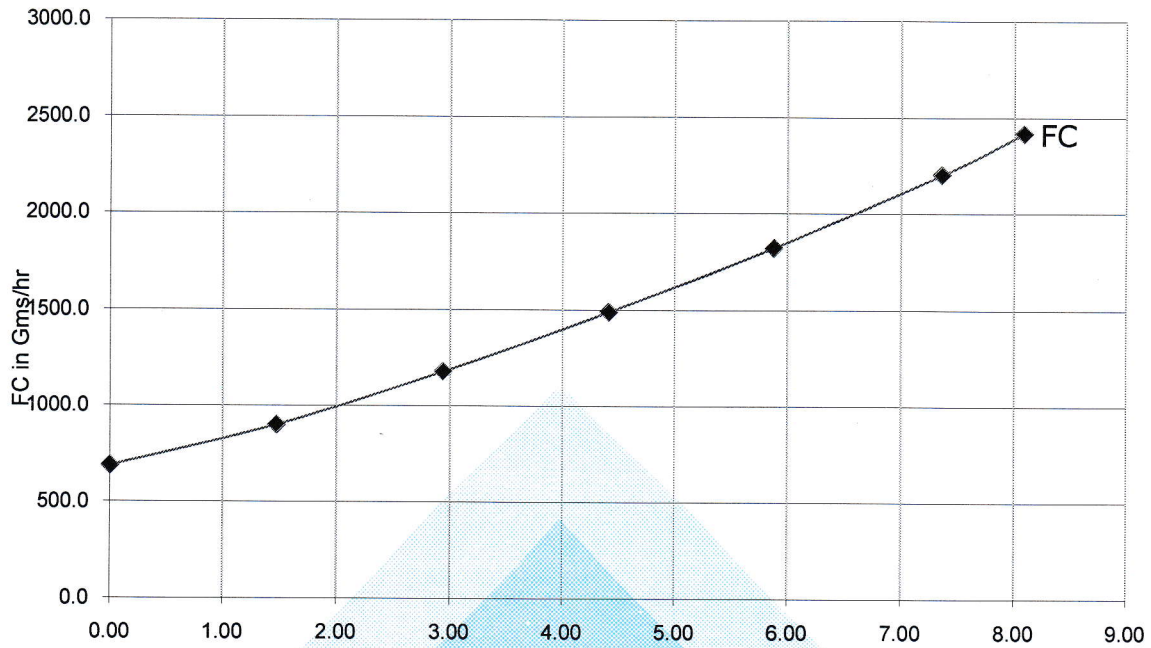
NIL

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ULR No. TC754425100000030P

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ENGINE-RATING TEST

Page no.: 6 of 6

Report No. NTSC-RJT/DETL/2025030 ULR No. TC754425100000030P
Reference : VISHWAKARMA INDUSTRIES IS: 11170 : 1985
Type : Vertical
Brand/Model: MRS 5103 (MARIN APPLICATION) BHP/KW : 10 | 7.35
Rated Speed : 3000 RPM C.C. : Lit.
Bore : mm
Nomonal Stroke : mm
Comp. Ratio : No. of Cyn. : 1
Fuel Specification : H.S.D.SP.GR. 0.82
Lube Oil Specification : SAE 20W40
Place of Test : RAJKOT

Sheet No. 1 of 1
Engine Started & Loaded at : 6.00 Hrs.
Engine Stopped at : 22.00 Hrs.
Initial Lube Oil Temp. : 298 K
Final Lube Oil Temp. : 358 K
Dyanometer (Type) : Eddy Current
Brake Pulley Diameter : - mtr.
Rope Diameter : - mtr.
Dyanometer Constant : 7024
Altitude : 140 Meters

TIME IN HOURS & MIN.	BAROMETER READING IN Kpa	AIR TEMPERATURE IN K			RELATIVE HUMIDITY IN %	EXHAUST BACK PRESSURE IN Kpa	CORRECTION FAC. FOR POWER	OBSERVED SPEED IN R.P.M.	BRAKE LOAD IN NM.	CALCULATED POWER IN KW	CORRECTED POWER IN KW	CORRECTION FAC. FOR FUEL	FUEL CONSUMPTION DATA				COOLING WATER TEMP. IN K		EXHAUST GAS TEMP. IN K
		WET BULB	DRY BULB	AT ENGINE INTAKE									TIME FOR 41 gm	F.C. IN g/h	S.F.C. g/kw-h	CORR. SFC g/kw-h	IN LET	OUT LET	
7.00	99.9	288	291	295	71	1.10	1.016	3000	23.80	7.47	7.35	0.998	66	2236	299.3	299.9	-	-	761
8.00	99.9	289	293	297	64	1.10	1.009	3000	23.64	7.42	7.36	0.999	66	2236	301.3	301.7	-	-	760
9.00	100.0	289	295	299	50	1.10	1.007	3000	23.58	7.40	7.35	0.999	66	2236	302.1	302.4	-	-	760
10.00	100.0	289	296	300	45	1.10	1.006	3000	23.56	7.40	7.35	0.999	67	2203	297.8	298.0	-	-	761
11.00	100.0	290	297	301	46	1.10	1.000	3000	23.42	7.35	7.35	1.000	67	2203	299.6	299.6	-	-	761
12.00	99.9	290	299	303	36	1.10	0.996	3000	23.32	7.32	7.35	1.001	67	2203	300.9	300.7	-	-	761
13.00	99.9	291	300	304	38	1.10	0.990	3000	23.18	7.28	7.35	1.002	67	2203	302.7	302.2	-	-	762
14.00	99.7	291	302	306	29	1.10	0.986	3000	23.10	7.25	7.36	1.002	66	2236	308.4	307.6	-	-	762
15.00	99.6	291	303	307	26	1.10	0.983	3000	23.05	7.24	7.36	1.003	66	2236	309.0	308.1	-	-	762
16.00	99.6	292	303	307	31	1.10	0.980	3000	22.95	7.21	7.35	1.003	66	2236	310.4	309.3	-	-	762
17.00	99.6	292	303	307	31	1.10	0.980	3000	22.95	7.21	7.35	1.003	67	2203	305.8	304.7	-	-	762
18.00	99.6	292	302	306	35	1.30	0.981	2987	25.40	7.94	8.09	1.003	61	2420	304.8	303.8	-	-	838
FUEL CONSUMPTION TEST START AT 18.30 Hrs. 28-01-2025																			
19.00	99.6	292	301	305	39	0.15	0.983	3181	0.00	0.00	0.00	1.003	215	687	-	-	-	-	503
19.30	99.6	291	300	304	38	0.30	0.987	3140	4.42	1.45	1.47	1.002	164	900	619.7	618.3	-	-	527
20.00	99.6	291	300	304	38	0.50	0.987	3122	8.88	2.90	2.94	1.002	125	1181	407.0	406.1	-	-	567
20.30	99.7	291	299	303	42	0.70	0.990	3070	13.60	4.37	4.41	1.002	99	1491	341.2	340.7	-	-	614
21.00	99.9	291	298	302	47	0.90	0.994	3041	18.36	5.84	5.88	1.001	81	1822	311.9	311.6	-	-	680
21.30	99.9	291	298	302	47	1.10	0.994	3000	23.30	7.31	7.36	1.001	67	2203	301.2	300.9	-	-	762
22.00	99.9	291	298	302	47	1.30	0.994	2987	25.72	8.04	8.09	1.001	61	2420	301.0	300.7	-	-	838

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End Of Report